Aritra Kundu

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Research Summary

My research focuses on understanding the first galaxies and their environment during the Epoch of Reionization (EoR). I approach this using both the present-day and high-redshift galaxies through cosmological simulations. Using the present-day intact and disrupted low-mass galaxies around Milky Way (MW)/Local Group (LG)-like host galaxies, I infer the properties of their ancient progenitors during EoR. I have generated multiple synthetic observations of these present-day low-mass galaxies for the Legacy Survey of Space and Time (LSST) and the Roman Space Telescope which are publicly available for interpreting the upcoming data from these instruments. I have also built semi-analytical models of Lyman-alpha (Ly-α) emission from the high-redshift galaxies in large-scale simulations to understand the interplay between these ionizing sources and their surrounding intergalactic medium (IGM) to interpret the recent and upcoming James Webb Space Telescope (JWST) and Roman observations during the EoR. By combining these two complementary approaches, I seek to answer the fundamental questions about the reionization history of our Universe - *When was the Universe reionized?*, *How was the Universe reionized?*, and *What were the sources of reionization?*

Research Interests

- First galaxies and reionization
- Near-field cosmology with both intact and disrupted present-day low-mass galaxies
- Synthetic observations for LSST, Roman, and JWST
- Theoretical models for Ly-α transmissions during EoR and predictions for cosmological cross-correlations.

Education

Ph.D., Physics & Astronomy, University of Pennsylvania — Philadelphia, USA
Advisors: Prof. Robyn Sanderson & Prof. Adam Lidz — 2022 – 2027
Thesis: *Probing Reionization using both Low-z and High-z Galaxies in Cosmological Simulations.*

M.Sc., Physics, Presidency University — Kolkata, India
Advisors: Prof. Ritaban Chatterjee — 2019 – 2021
Thesis: *RMS-Flux Relation and Disc-Jet Connection in Blazars in the context of the Internal Shocks Model*

B.Sc., Physics, Presidency University — Kolkata, India
Advisors: Prof. Kanan K. Datta — 2016 – 2019
Thesis: *Cosmic Recombination History in Light of EDGES Measurements of the Cosmic Dawn 21 cm Signal*

Publications and Preprints

Primary Author

1. **Kundu**, **A.**, et al. (2026), "Peering into reionization with present-day LSST-detectable stellar streams and satellite galaxies (in prep.),"

2. **Kundu**, **A.**, Lidz, A. (2026), "Probing the Reionization History and Bubble Sizes with JWST Lyman-α Fraction Measurements (in prep.),"

3. **Kundu**, **A.**, Sanderson, R., Lidz, A., Gandhi, P. J., Wetzel, A., Feldmann, R., et al. (2026), "Rise of the Forsaken Relics: Connecting Present-day Stellar Streams and Phase-mixed Galaxies to the Epoch of Reionization," ApJ, 999, 124.

4. **Kundu**, **A.**, Chatterjee, R., Mitra, K., Mondal, S. (2022), "rms-flux relation and disc-jet connection in blazars in the context of the internal shocks model," MNRAS, 510, 3688.

5. Datta, K. K., **Kundu**, **A.**, Paul, A., Bera, A. (2020), "Cosmic recombination history in light of EDGES measurements of the cosmic dawn 21-cm signal," PRD, 102, 083502.

Co-Authored

1. Mandacarú Guerra, L. S., O'Neil, S., Lisanti, M., Roy, S., Sanderson, R., **Kundu**, **A.**, et al. (2026), "Probing Atomic Dark Matter with Stellar Streams in Milky Way-Mass Galaxies," arXiv e-prints, arXiv:2603.20367.

2. Tasso, E., Lucey, M., Sanderson, R., **Kundu**, **A.**, Necib, L. (2026), "The dynamics of old inner galaxy stars in Milky Way-mass galaxies using FIRE-2 simulations," MNRAS, 550, stag1139.

3. Lucey, M., Sanderson, R. E., Horta, D., **Kundu**, **A.**, Hopkins, P. F., Arora, A., et al. (2025), "Cosmological Predictions for Minor Axis Stellar Density Profiles in the Inner Regions of Milky Way–mass Galaxies," ApJ, 982, 87.

Research Presentations

Colloquia & Seminars

1. **Near-field Cosmology group meeting**, *'Probing Reionization using Low-z & High-z Galaxies'*, **Aug 2026, University of Washington**, Seattle, USA

2. **Stellar Streams group meeting**, *'Probing Reionization using Low-z & High-z Galaxies'*, **July 2026, Carnegie Observatories**, Pasadena, USA

3. **UC Davis Physics Special Seminar**, *'Probing Reionization using Low-z & High-z Galaxies'*, **July 2026, UC Davis**, Davis, USA

Conferences & Workshops

1. **Contributed Talk**, *'Roman Lyman-Alpha Emitters at Cosmic Dawn and the Prospects for LIM Cross-Correlations'*, Aug 2026, **Line Intensity Mapping 2026**, California Institute of Technology, Pasadena, USA

2. **Contributed Talk**, *'Rise of the Forsaken Relics: Probing reionization using disrupted galaxies'*, Jul 2026, **Near-field Cosmology in the Era of Big Data: Local Group and Beyond**, Fields Institute, Toronto, Canada

3. **Contributed Talk**, *'Rise of the Forsaken Relics: Probing reionization using disrupted galaxies'*, Jul 2026, **Shifting Landscapes in Astrophysics: New Frontiers to Explore with Roman**, California Institute of Technology, Pasadena, USA

4. **Contributed Poster**, *'Probing reionization with disrupted substructures'*, Jul 2024, **Small Galaxies, Cosmic Questions - II**, Durham University, Durham, UK

5. **Contributed Poster**, *'RMS-Flux relation and disc-jet connection in blazars in the context of the internal shocks model'*, Mar 2022, **40th Meeting of the Astronomical Society of India**, IIT

Roorkee & ARIES Nainital, India (virtual)

6. **Contributed Talk**, *'RMS-Flux relation and disc-jet connection in blazars in the context of the internal shocks model'*, Feb 2022, **21st National Space Science Symposium,**, India (virtual)
7. **Contributed Talk**, *'Cosmic recombination history in light of EDGES measurements of the cosmic dawn 21-cm signal'*, Oct 2021, **4th Global 21 cm workshop**, University of Colorado, Boulder, USA (virtual)
8. **Contributed Talk**, *'Cosmic recombination history in light of EDGES measurements of the cosmic dawn 21-cm signal'*, Feb 2021, **39th Meeting of the Astronomical Society of India**, ICTS-TIFR Bengaluru, IISER Mohali, IIT Indore & IUCAA Pune, India (virtual)
9. **Contributed Poster**, *'Impact of colder IGM scenario on the Recombination History of the Universe'*, January 2020, **Observing the First Billion Years of the Universe**, IIT Indore, India

Awards and Fellowships

1. Received the Parthasarathi Gupta Memorial Fund, for being first in M.Sc. Physics 2021
2. Gold medalist in M.Sc. Physics 2021
3. Received the Satyabrata Ghosh Cash Prize 2020 in B.Sc. for highest grades in Physics in 2019

Mentoring Experience

Sara Michele Alvarado Rodriguez, *UPenn undergraduate* 2025 - present
Penn Undergraduate Research Mentoring Program
– Primary mentor on *Cosmological predictions for stellar populations in LSST-detectable tidal streams.*
– Paper to be submitted soon.

Alexandra DiMauro, *UPenn undergraduate* 2024 - present
Penn Undergraduate Research Mentoring Program
– Primary mentor on *Signatures of CDM and SIDM Models in Dwarf Galaxies in FIRE-2 Simulations.*
– Paper to be submitted soon.

Teaching Experience

Guest Lectures

University of Pennsylvania

ASTR 1212, Introduction to Astrophysics II Spring 2025, 2024
– Lectured on reionization

ASTR 1211, Introduction to Astrophysics I Fall 2025
– Taught a class on Doppler effect

Teaching Assistantships

University of Pennsylvania

ASTR 1211, Introduction to Astrophysics I — Fall 2026, 2025
– Helped with Python-lab exercises, grading, office hours, tutoring, exam reviews.

ASTR 3392, Life & Death of Stars — Spring 2026
– Helped with grading.

ASTR 1212, Introduction to Astrophysics II — Spring 2025, 2024, Fall 2022
– Helped with Python-lab exercises, grading, office hours, tutoring, exam reviews.

ASTR 0007, The Big Bang & Beyond — Fall 2024
– Helped with grading, office hours.

ASTR 0001, The Survey of the Universe — Spring 2023
– Helped with grading.

Professional Activities and Outreach

1. Peer mentoring first-year graduate students at University of Pennsylvania from 2026-2027, to help them navigate the graduate school.
2. Organizer of the Departmental Friday Astro Lunch/Journal Club at the University of Pennsylvania since Fall 2024.
3. Member of Scientific Organizing Committee, Presision - Undergraduate Physics Symposium, in 2019 and 2021, Presidency University, Kolkata.
4. Co-organizer, Presision - Undergraduate Physics Symposium, in 2017, 2018 and 2020, Presidency University, Kolkata.

Skills

Programming Languages	Python (proficient), Fortran (intermediate), SQL (basic)
Languages	Bengali (native), English (fluent), Hindi (fluent)